

EnerZen Performance Engine — Methodology

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This document describes every calculation the engine performs, at the level of the underlying formulas and constants. It is written to be auditable: each section names the source file so a reader can check the maths against the code.

1. What the engine does

The engine takes a project description and searches for the best assembly configuration. It is a deterministic physics-and-costing model — there is no machine learning anywhere in it.

The search is exhaustive. The catalog holds 5 wall panels, 3 roof cassettes, 2 floor cassettes, 3 window packages and 3 mechanical systems, so the engine evaluates every combination:

$5 \times 3 \times 2 \times 3 \times 3 = 270$ candidate configurations

For each candidate it runs five independent calculation modules, discards any that breach the budget (or the performance target), and ranks the survivors.

```
Project inputs
|
v
For each of 270 combinations:
|-- simulator.py : energy demand, EUI, EnerGuide, NZR probability
|-- cost.py      : construction cost, build schedule
|-- carbon.py    : embodied + operational carbon
|-- solar.py     : PV generation, cost, carbon
|-- finance.py   : monthly utility bill, 60-year lifecycle cost
|
v
Filter (budget, target label)
|
v
Rank (Pareto dominance, then weighted score)
|
v
Ranked list of configurations
```

Inputs

Input	Meaning
Building type	single family, townhouse, or MURB
Floor area	conditioned floor area, m2
Storeys	1, 2 or 3 — selects the surface-area ratios
Climate zone	6, 7a or 7b — selects degree days and thresholds
Orientation	main facade direction; drives solar gain and PV yield

Input	Meaning
Window-to-wall ratio	fraction of wall area that is glazing
Performance target	code / NZR / Passive House — sets the airtightness target
Solar option	rooftop PV array size
Budget per unit	hard constraint; configurations above it are discarded
Priority weights	cost / speed / carbon / energy, used for final ranking

Outputs

Optimized assembly selection, construction cost, build time, embodied carbon, operational energy, EnerGuide estimate, Net Zero Ready probability, lifecycle cost, and a monthly utility estimate.

2. Energy model

Source: `engine/simulator.py`

A steady-state **degree-day heat balance**. The governing idea: heat escapes a building in proportion to how leaky its surfaces are multiplied by how cold it is outside, over the whole heating season.

This is a simplified stand-in for HOT2000. The module is designed so it can be replaced by a HOT2000 wrapper without changing its interface.

2.1 Surface areas

Areas are derived from floor area using fixed ratios per storey count. These are generic residential forms, not EnerZen's actual designed units.

Storeys	Wall ratio	Roof ratio	Floor ratio
1	1.80	1.05	1.00
2	1.40	0.55	0.52
3	1.20	0.40	0.38

```

wall_area      = floor_area x wall_ratio
roof_area      = floor_area x roof_ratio
floor_area_surf = floor_area x floor_ratio
window_area    = wall_area x window_to_wall_ratio
opaque_wall_area = wall_area - window_area

```

2.2 Heat loss coefficient (UA)

UA is the rate of heat loss per degree of temperature difference, in watts per kelvin (W/K). For each opaque or glazed surface it is area multiplied by the assembly U-value (U is the inverse of R — lower U means better insulation).

```

UA_wall  = opaque_wall_area x wall_U
UA_roof  = roof_area        x roof_U

```

```

UA_floor    = floor_area_surf x floor_U
UA_windows  = window_area      x window_U

```

Air leakage is handled separately. A blower-door test reports ACH50 (air changes per hour at 50 pascals of pressure). Real-world infiltration is far lower than the test condition; the model uses the standard **divide-by-20 rule of thumb**:

```

ACH_natural    = ACH50 / 20
volume         = floor_area x 2.7 x storeys
UA_infiltration = ACH_natural x volume x 0.33

```

- 2.7 is the assumed storey height in metres.
- 0.33 is the volumetric heat capacity of air in Wh/(m3.K).

A heat recovery ventilator (HRV) reclaims heat from exhaust air, so only the unrecovered fraction counts:

```

UA_vent = UA_infiltration x (1 - hrv_efficiency)
UA_total = UA_wall + UA_roof + UA_floor + UA_windows + UA_vent

```

2.3 Heating demand

```

heating = UA_total x HDD x 24 / 1000 x solar_factor / COP

```

- **HDD** — heating degree days for the climate zone (see section 8). It aggregates how cold the year is.
- **x 24 / 1000** — converts watt-degree-days into kilowatt-hours.
- **solar_factor** — free solar heat gain reduces the heating that must be purchased. A south-facing home benefits most.

Orientation	Solar factor
South	0.82
East	0.90
West	0.90
North	0.95

- **COP** — coefficient of performance of the heating plant. A heat pump with a COP of 2.5 delivers 2.5 kWh of heat per kWh of electricity, so dividing by COP converts heat *demand* into *purchased energy*. A gas furnace has a COP below 1 (0.92), reflecting combustion losses.

2.4 Cooling demand

Cooling is modelled as solar gain through glazing only, which is a heavy simplification appropriate to the modest Canadian cooling season.

```

cooling = window_area x window_SHGC x CDD x 24 / 1000 x 0.4 / 3.5

```

- **SHGC** — solar heat gain coefficient of the glazing.
- **CDD** — cooling degree days.
- **0.4** — an empirical factor accounting for the fraction of gain that actually drives mechanical cooling.
- **3.5** — assumed cooling COP (hard-coded, not taken from the mechanical entry).

2.5 Base loads

```
hot_water = 15 x floor_area / storeys
appliances = 25 x floor_area
```

Both in kWh/yr. Appliance and lighting load is a flat intensity and does not vary with the assembly.

2.6 EUI and EnerGuide

```
total_energy = heating + cooling + hot_water + appliances
EUI = total_energy / floor_area
```

EUI (energy use intensity, kWh/m²/yr) is the headline efficiency figure and the basis of the Net Zero Ready test.

The EnerGuide score is a linear approximation of EUI, clamped to 0-100:

```
EnerGuide = clamp(0, 100, 100 - (EUI - 30) x 0.8)
```

This is an estimate for guidance only, not an official rating.

2.7 Net Zero Ready probability

A single deterministic EUI hides the fact that real buildings vary. Two homes that both model at "compliant" are not equally safe if one sits at the threshold and the other well below it.

The engine therefore runs a **Monte Carlo simulation**: it re-runs the energy model 400 times, each time drawing the uncertain real-world inputs from a normal distribution, and reports the fraction of runs that meet the threshold.

Sampled input	Distribution	Represents
Weather severity	Normal(1.00, 0.06)	year-to-year weather variation
As-built airtightness	Normal(1.00, 0.20)	blower-door scatter vs. target
Occupant plug loads	Normal(1.00, 0.20)	occupant behaviour
Mechanical COP derate	Normal(0.97, 0.05)	real-world vs. rated efficiency

Each sample is floored at a physical minimum to avoid nonsensical draws. The run is seeded, so results are stable between runs.

```
NZR probability = (runs with EUI <= threshold) / 400
```

A home with margin scores near 100 percent; one sitting on the threshold scores near 50 percent. The probability is envelope-based and deliberately excludes PV, matching the CHBA definition where Net Zero *Ready* describes the building itself, not its solar array.

3. Construction cost

Source: engine/cost.py

```
total = materials + labour + base_cost
```

3.1 Materials

Each surface area multiplied by its catalog unit rate, plus the flat mechanical system cost.

```
materials = opaque_wall_area x wall_cost_per_m2
           + roof_area       x roof_cost_per_m2
           + floor_area_surf x floor_cost_per_m2
           + window_area     x window_cost_per_m2
           + mechanical_cost
```

3.2 Labour

Only wall, roof and floor installation hours are counted. Window and mechanical installation labour is **not** modelled.

```
labour_hours = opaque_wall_area x wall_install_hours_per_m2
               + roof_area       x roof_install_hours_per_m2
               + floor_area_surf x floor_install_hours_per_m2

labour = labour_hours x 75
```

The blended crew rate is **75 CAD/hour**. This is where panelized assemblies pay off: they cost more per square metre in materials but need roughly half the installation hours.

3.3 Base cost

```
base_cost = floor_area x 1200
```

A flat **1200 CAD/m2** covering everything outside the envelope model: foundation, structural framing, interior finishes, plumbing, electrical.

It is deliberately held constant across configurations so that comparing two assemblies only moves the parts that genuinely differ. The consequence is that roughly two thirds of the reported cost is a placeholder rather than a modelled quantity, and assembly choice only swings the remaining third.

3.4 Solar

PV is added on top of the construction cost in the optimizer:

```
total_cost = construction_cost + pv_capacity_kw x pv_cost_per_kw
```

3.5 Worked example

A 150 m2, two-storey home, 20 percent window-to-wall, with wall W4, roof R2, floor F2, windows GL2 and mechanical M3.

```
wall_area      = 150 x 1.40 = 210.0 m2
window_area    = 210 x 0.20 = 42.0 m2
opaque_wall_area = 210 - 42 = 168.0 m2
roof_area      = 150 x 0.55 = 82.5 m2
floor_area_surf = 150 x 0.52 = 78.0 m2
```

Component	Quantity	Rate	Cost
Opaque wall (W4)	168.0 m2	155 /m2	26,040
Roof (R2)	82.5 m2	110 /m2	9,075
Floor (F2)	78.0 m2	95 /m2	7,410
Windows (GL2)	42.0 m2	680 /m2	28,560

Component	Quantity	Rate	Cost
Mechanical (M3)	1	20,000	20,000
Materials subtotal			91,085
Labour	135.9 hr	75 /hr	10,193
Base cost	150 m2	1,200 /m2	180,000
Total			281,278

4. Build schedule

Source: engine/cost.py

```
weeks = labour_hours / (4 x 40)
```

A four-person crew working a forty-hour week gives 160 labour-hours per week. Labour hours are exactly those from the cost model.

Using the worked example above, 135.9 hours gives roughly 0.85 weeks. The same home in stick-frame assemblies (W1/R1/F1) needs about 312.6 hours, or roughly 2.0 weeks — about 2.3 times slower.

Panel counts assume a standard 2.4 m x 3.0 m panel:

```
panel_area    = 2.4 x 3.0 = 7.2 m2
wall_panels   = round(opaque_wall_area / panel_area)
roof_panels   = round(roof_area / panel_area)
floor_panels  = round(floor_area_surf / panel_area)
crane_lifts   = wall_panels + roof_panels + floor_panels
```

Scope limitation. This figure is time to *envelope close* only. Foundation, window and mechanical installation, interior fit-out, inspections and cure times are excluded, as is any allowance for weather, staging or crane scheduling. It is not a project programme.

5. Carbon

Source: engine/carbon.py

5.1 Embodied carbon

Carbon emitted producing the materials, from EPD-based values in the catalog.

```
embodied = opaque_wall_area x wall_embodied_per_m2
          + roof_area       x roof_embodied_per_m2
          + floor_area_surf x floor_embodied_per_m2
          + window_area     x window_embodied_per_m2
          + mechanical_embodied
```

PV embodied carbon is added in the optimizer at 1500 kgCO₂e per kW installed.

The module also reports a **carbon hotspot** — whichever single component contributes the most embodied carbon.

5.2 Operational carbon

```
grid_factor      = 0.19 if mechanical is gas else 0.074
operational_annual = total_energy x grid_factor
operational_60yr  = operational_annual x 60
```

- **0.074 kgCO₂e/kWh** — Ontario grid intensity, 2024.
- **0.19 kgCO₂e/kWh** — natural gas combustion.

Note that Ontario's grid intensity rose 25 percent in 2024 as gas-fired generation increased. An earlier version of this engine used 0.03, a 2022-era figure, which understated the operational carbon of electric systems by roughly two and a half times.

6. Solar

Source: engine/solar.py

A specific-yield model. Annual output is the array size multiplied by the yield per installed kilowatt for the region, derated for orientation.

```
generation = capacity_kw x specific_yield x orientation_derate
cost        = capacity_kw x cost_per_kw
embodied    = capacity_kw x embodied_per_kw
```

Orientation	Derate
South	1.00
East	0.90
West	0.90
North	0.80

The orientation derate uses the main facade as a proxy for the available roof plane, which is an approximation.

Net energy and net zero

```
net_operational = total_energy - generation
net_EUI         = net_operational / floor_area
net_zero        = net_operational <= 0
```

Net Zero Ready and **Net Zero** are different tests. Net Zero Ready concerns the envelope and is assessed before PV. Net Zero requires generation to meet or exceed consumption over the year.

7. Utility bill and lifecycle cost

Source: engine/finance.py

7.1 Monthly utility bill

Annual demand is distributed across the year using typical Ontario monthly profiles, then priced. Each profile is normalised so its twelve values sum to 1.

- **Heating** follows the heating degree-day share (peaks in January).
- **Cooling** follows the cooling degree-day share (peaks in July).
- **Base loads** — hot water, appliances, lighting — are spread evenly.
- **PV generation** follows the irradiance share (peaks in summer).

For each month:

```
base_monthly = (total_energy - heating - cooling) / 12
elec_kwh      = cooling_share + base_monthly + (heating if plant is electric)
gas_kwh       = heating_share if plant is gas else 0
net_elec      = elec_kwh - pv_generation_share
elec_cost     = max(net_elec, 0) x electricity_rate
gas_cost      = gas_kwh x gas_rate
```

Net metering is modelled within the month, and the bill floors at zero: surplus generation offsets consumption but is not paid out or banked across months.

7.2 Lifecycle cost

A 60-year present-value calculation.

```
upfront = construction_cost + pv_cost - solar_rebate

for each year y in 1..60:
    escalated = annual_energy_cost x (1 + 0.02)^(y-1)
    pv_energy = pv_energy + escalated / (1 + 0.03)^y

lifecycle_total = upfront + pv_energy
```

Parameter	Value
Study period	60 years
Discount rate	3 percent
Energy price escalation	2 percent per year

Maintenance, component replacement (a heat pump will not last 60 years) and residual value are **not** modelled.

8. Optimizer and ranking

Source: engine/optimizer.py

8.1 Airtightness by target

The performance target sets the blower-door target used by the energy model.

Target	ACH50
Code	5.0

Target	ACH50
Net Zero Ready	3.0
Passive House	1.5

8.2 Filtering

A configuration is discarded if its total cost (envelope plus PV) exceeds the budget. If the target is Net Zero Ready, configurations whose deterministic EUI misses the threshold are also discarded.

8.3 Pareto ranking

Configuration B **dominates** A when B is at least as good as A on all four objectives — cost, build time, embodied carbon and EUI — and strictly better on at least one.

$$\text{rank}(A) = 1 + \text{count of configurations that dominate } A$$

Rank 1 configurations are non-dominated: nothing else beats them on every axis simultaneously. They represent the genuine trade-off frontier.

8.4 Weighted score

Within a rank, objectives are min-max normalised across all feasible configurations to a 0-1 scale, then combined using the priority weights. Lower is better on every objective, so a lower score is better.

$$\begin{aligned} \text{norm}(v) &= (v - \min) / (\max - \min) \\ \text{score} &= w_{\text{cost}} \times \text{norm}(\text{cost}) \\ &\quad + w_{\text{speed}} \times \text{norm}(\text{weeks}) \\ &\quad + w_{\text{carbon}} \times \text{norm}(\text{embodied_carbon}) \\ &\quad + w_{\text{energy}} \times \text{norm}(\text{EUI}) \end{aligned}$$

Results are sorted by Pareto rank first, then by weighted score. The top result is presented as the recommended configuration.

9. Reference data

Climate zones

Zone	Region	HDD	CDD	Design temp	NZR EUI threshold	Solar yield
6	Toronto / Southern Ontario	3520	320	-18 C	60 kWh/m2/yr	1250 kWh/kWp
7a	Ottawa / Eastern Ontario	4440	220	-25 C	55 kWh/m2/yr	1200 kWh/kWp
7b	Northern Ontario / Sudbury	5500	130	-30 C	50 kWh/m2/yr	1150 kWh/kWp

Energy rates and factors

Parameter	Value	Source
Electricity	0.16 CAD/kWh	OEB TOU/tiered ~12c/kWh energy + delivery/regulatory; blended all-in Nov 2025
Natural gas	0.055 CAD/kWh	Enbridge Ontario all-in ~\$0.50/m ³ / 10.35 kWh per m ³
Grid intensity	0.074 kgCO ₂ e/kWh	The Atmospheric Fund, Ontario Emissions Factors 2024 (73.8 gCO ₂ e/kWh)
Solar rebate	5000 CAD	Ontario Home Renovation Savings Program (HRS), ~\$5k toward rooftop PV

Comparison benchmarks

Metric	Value	Source
Existing home EUI	200 kWh /m ² /yr	NRCan residential end-use intensity; CHBA Net Zero program
Code-built new home EUI	130 kWh /m ² /yr	NRCan residential end-use intensity; CHBA Net Zero program
Conventional build cost	2200 CAD/m ²	Ontario custom-build cost surveys 2025 (~\$205/sqft avg)
Conventional embodied carbon	184 kgC O ₂ e/m ²	Cradle-to-gate mean for new low-rise residential (Living Materials Lab / ScienceDirect, 2024)

10. Assembly catalog

Wall panels

ID	Name	R	U (W/m ² K)	Cost /m ²	Carbon /m ²	Install hr/m ²
W1	2x6 Wood Frame + R22 Batt	R22	0.045	85	18.5	0.9
W2	2x6 Wood Frame + R22 Batt + R10 Exterior Mineral Wool	R32	0.031	120	22.1	1.1
W3	Panelized SIP Wall (R28)	R28	0.036	140	28.4	0.5
W4	EnerZen Panelized Wall — Standard (R40)	R40	0.025	155	24.8	0.45
W5	EnerZen Panelized Wall — High Performance (R50)	R50	0.02	185	27.3	0.45

Roof cassettes

ID	Name	R	U (W/m2K)	Cost /m2	Carbon /m2	Install hr/m2
R1	Conventional Attic (R40)	R40	0.025	65	12.0	1.2
R2	EnerZen Roof Cassette (R60)	R60	0.017	110	15.8	0.4
R3	EnerZen Roof Cassette (R80)	R80	0.012	145	18.2	0.4

Floor cassettes

ID	Name	R	U (W/m2K)	Cost /m2	Carbon /m2	Install hr/m2
F1	Slab on Grade (R10 under)	R10	0.1	55	32.0	0.8
F2	EnerZen Floor Cassette (R30)	R30	0.033	95	14.5	0.35

Window packages

ID	Name	U (W/m2K)	SHGC	Cost /m2	Carbon /m2
GL1	Double-Pane Low-E (U=1.8)	1.8	0.4	420	65.0
GL2	Triple-Pane Low-E (U=0.8)	0.8	0.35	680	82.0
GL3	Triple-Pane Passive House (U=0.6)	0.6	0.5	820	88.0

Mechanical systems

ID	Name	Type	Heating COP	Cooling COP	HRV	Cost	Carbon
M1	Gas Furnace + Central AC	gas	0.92	3.0	-	12000	850
M2	Cold Climate Air Source Heat Pump	electric	2.5	3.5	-	16000	650
M3	Cold Climate ASHP + HRV	electric	2.5	3.5	0.8	20000	720

Solar options

ID	Name	Capacity	Cost /kW	Carbon /kW
PV0	No Solar	0 kW	0	0
PV1	Rooftop PV — Small (4 kW)	4 kW	2900	1500
PV2	Rooftop PV — Standard (8 kW)	8 kW	2700	1500
PV3	Rooftop PV — Large (12 kW)	12 kW	2500	1500

11. Assumptions and limitations

These are the things a reader should know before trusting a number from this tool.

The energy model is simplified. A steady-state degree-day method ignores thermal mass, hourly weather, real shading geometry, part-load equipment behaviour and distribution losses. It is not HOT2000 and is not a compliance tool. Reported EUIs run optimistic relative to published benchmarks; part of that gap is genuine performance and part is model simplification.

Geometry is generic. Surface areas come from floor-area ratios for typical residential forms, not from EnerZen's actual designed units. Storey height is assumed at 2.7 m for the volume calculation, while wall area comes from the ratios — so height is effectively assumed twice, by two unlinked routes.

Cost is dominated by a placeholder. The 1200 CAD/m² base cost is roughly two thirds of the reported total and is not modelled. Window and mechanical installation labour is omitted entirely.

Build time covers envelope close only. It is not a project schedule.

Cooling is glazing-driven only, and uses a hard-coded COP of 3.5 rather than the selected mechanical system's rating.

Lifecycle cost omits maintenance and replacement, which will understate the true cost of any system over 60 years.

Benchmarks and rates are provincial averages drawn from secondary sources. They are adequate for directional comparison in a prototype and should be replaced with EnerZen's own project data before any number is quoted to a client.

The Monte Carlo distributions are engineering judgement. They are plausible representations of as-built variance but are not calibrated against measured EnerZen blower-door or energy-bill data. The probability is therefore a structured estimate of confidence, not a validated statistic.